

Related Work for Cone-Aware Primal-Dual Graph Unrolling

Abstract

This note summarizes related work on sparse optimization, graph neural networks for mathematical programming, and algorithm-unrolled learning-to-optimize methods. The literature contains mature unrolled solvers for LASSO-type sparse recovery, WL-based GNN expressivity results for LP/QP/QCQP/SOCP, and PDHG-unrolled architectures for LPs and QPs. The remaining gap is a cone-aware, dimension-agnostic primal-dual graph architecture that unrolls SOCP first-order iterations while retaining explicit second-order cone projections.

1 Related Work

GNN expressivity for learning to optimize. Graph neural networks have become a common inductive bias for mathematical programs whose variables and constraints admit permutation-equivariant representations. The WL-based line of work represents an optimization instance as a graph and relates the expressive power of message passing to Weisfeiler–Lehman graph tests. For LPs, Chen et al. [6] represent variables, constraints, and coefficients by a weighted bipartite graph and establish approximation results for LP feasibility, boundedness, optimal values, and optimal solutions. This framework has been extended to quadratic and mixed-integer quadratic programs [7], as well as convex quadratically constrained quadratic programs [19]. More general provable GNN solvers for convex optimization have also been studied [16]. Recent SOCP-GNN results extend WL-based expressivity to second-order cone constraints by augmenting the graph representation to encode conic blocks and prove universality and sample-complexity guarantees [11, 12]. These results show that SOCPs can be represented by suitably designed GNNs, but they do not by themselves specify an algorithmic trajectory, a residual-driven primal-dual loss, or an explicit solver warm-start procedure. Complementary negative results for SDPs further indicate that graph architectures must match the algebraic structure of the underlying cone; standard message passing may fail when the cone geometry introduces higher-order symmetries [17].

Algorithm-unrolled learning to optimize. Algorithm unrolling maps iterations of an optimization algorithm to neural network layers, preserving useful solver structure while replacing hand-designed parameters by trainable ones [15]. The primal-dual hybrid gradient method [2] is especially relevant

because it solves saddle-point formulations of the form

$$\begin{aligned} & \min_x f(x) + g(Kx) \\ \iff & \min_x \max_y f(x) + \langle Kx, y \rangle - g^*(y), \end{aligned} \quad (1)$$

PDHG-Net unrolls this structure for large-scale LPs and uses channel expansion to avoid dense trainable matrices tied to a fixed problem size [10]. PDQP-Net adopts a related primal-dual unrolling strategy for convex QPs and trains with KKT-informed unsupervised losses [21]. Other constrained unrolling approaches use coupled primal and dual GNNs to emulate dual-ascent dynamics for general constrained problems [9], while doubly stochastic PDHG studies blockwise primal-dual updates and sampling mechanisms in classical first-order optimization [20]. This AU-based paradigm provides algorithmic interpretability and size generalization, but extending it from LP/QP updates to conic programs requires representing non-linear cone projections and cone-block interactions inside the learned architecture.

Sparse optimization and graph unrolling. Sparse recovery provides a canonical example of successful unrolling. The LASSO [18] solves

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1, \quad (2)$$

and proximal-gradient methods such as ISTA and FISTA exploit the soft-thresholding proximal operator of the ℓ_1 norm [1]. LISTA unrolls ISTA into a trainable architecture for sparse coding [8], and subsequent work established convergence guarantees and efficient hyperparameterized variants [4, 5]. Graph unrolling networks extend the same idea to graph signal denoising and include a graph sparse-coding architecture, whose objective can be written as a LASSO-type problem with a graph-filter dictionary [3]. Thus, sparse optimization already has mature neural and graph-unrolled variants. The unresolved LASSO setting considered here is not sparse coding on a fixed graph dictionary, but graph-parametric optimization over arbitrary sensing matrices, where the matrix A itself defines a variable-measurement bipartite graph and the learned update rule must generalize across dimensions.

Conic optimization and SOCP first-order methods. SOCPs extend LPs and convex QPs by introducing constraints of the form

$$\begin{aligned} & \min_x c^\top x \\ \text{s.t.} \quad & Fx \leq g, \quad l \leq x \leq r, \\ & \|A_i x + b_i\|_2 \leq d_i^\top x + e_i, \quad i = 1, \dots, q. \end{aligned} \quad (3)$$

where $\|A_i x + b_i\|_2 \leq d_i^\top x + e_i$ is equivalent to membership in the second-order cone $\mathcal{K}_{\text{soc}} = \{(t, z) : \|z\|_2 \leq t\}$ [14]. Classical primal-dual and splitting methods handle such constraints through projections onto products of cones. Recent large-scale conic solvers build on restarted PDHG, adaptive step-size rules, GPU-accelerated sparse matrix-vector products, and efficient projections onto zero, nonnegative, second-order, rotated second-order, exponential, and semidefinite cones [13]. These solvers establish that PDHG is a practical algorithmic backbone for conic programs. However, they remain hand-designed first-order methods rather than learned, dimension-agnostic graph architectures.

Scope of the present work. The closest learning-based SOCP work is WL-based: it proves that specialized GNNs can represent SOCP feasibility and solutions [12]. The closest unrolled learning-to-optimize works are AU-based: they unroll PDHG/PDQP for LPs and QPs using channel expansion and primal-dual residual losses [10, 21]. The closest classical solver work applies PDHG directly to conic programs with explicit cone projections [13]. To our knowledge, existing work has not combined these three ingredients into a cone-aware, PDHG-unrolled GNN for SOCPs that retains explicit second-order cone projections while learning step sizes, extrapolation, block-sampling probabilities, or channel-mixing maps.

References

- [1] Amir Beck and Marc Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009. doi: 10.1137/080716542.
- [2] Antonin Chambolle and Thomas Pock. A first-order primal-dual algorithm for convex problems with applications to imaging. *Journal of Mathematical Imaging and Vision*, 40(1):120–145, 2011. doi: 10.1007/s10851-010-0251-1.
- [3] Siheng Chen, Yonina C. Eldar, and Lingxiao Zhao. Graph unrolling networks: Interpretable neural networks for graph signal denoising. *IEEE Transactions on Signal Processing*, 69:3699–3713, 2021. doi: 10.1109/TSP.2021.3087905.
- [4] Xiaohan Chen, Jialin Liu, Zhangyang Wang, and Wotao Yin. Theoretical linear convergence of unfolded ista and its practical weights and thresholds. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [5] Xiaohan Chen, Jialin Liu, Zhangyang Wang, and Wotao Yin. Hyperparameter tuning is all you need for lista. In *Advances in Neural Information Processing Systems*, volume 34, pages 11678–11689, 2021.
- [6] Ziang Chen, Jialin Liu, Xinshang Wang, Jianfeng Lu, and Wotao Yin. On representing linear programs by graph neural networks, 2022.
- [7] Ziang Chen, Xiaohan Chen, Jialin Liu, Xinshang Wang, and Wotao Yin. Expressive power of graph neural networks for (mixed-integer) quadratic programs, 2024.
- [8] Karol Gregor and Yann LeCun. Learning fast approximations of sparse coding. In *Proceedings of the 27th International Conference on Machine Learning*, pages 399–406, 2010.
- [9] Samar Hadou and Alejandro Ribeiro. Unrolled graph neural networks for constrained optimization, 2025.
- [10] Bingheng Li, Linxin Yang, Yupeng Chen, Senmiao Wang, Qian Chen, Haitao Mao, Yao Ma, Akang Wang, Tian Ding, Jiliang Tang, and Ruoyu Sun. Pdhg-unrolled learning-to-optimize method for large-scale linear programming. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. doi: 10.48550/arXiv.2406.01908.
- [11] Ruizhe Li, Enming Liang, and Minghua Chen. On the expressivity of gnn for solving second order cone programming. In *NeurIPS Workshop on GPU-Accelerated and Scalable Optimization*, 2025.
- [12] Ruizhe Li, Enming Liang, and Minghua Chen. On the universality and complexity of gnn for solving second-order cone programs. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=wFttcDu6Fr>.
- [13] Zhenwei Lin, Zikai Xiong, Dongdong Ge, and Yinyu Ye. A practical gpu-enhanced matrix-free primal-dual method for large-scale conic programs, 2026.
- [14] Miguel Sousa Lobo, Lieven Vandenbergh, Stephen Boyd, and Hervé Lebret. Applications of second-order cone programming. *Linear Algebra and its Applications*, 284(1–3): 193–228, 1998. doi: 10.1016/S0024-3795(98)10032-0.
- [15] Vishal Monga, Yuelong Li, and Yonina C. Eldar. Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Processing Magazine*, 38(2):18–44, 2021. doi: 10.1109/MSP.2020.3016905.
- [16] Chendi Qian and Christopher Morris. Towards graph neural networks for provably solving convex optimization problems, 2025.
- [17] Chendi Qian and Christopher Morris. On the expressive power of gnns to solve linear sdps, 2026.
- [18] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.
- [19] Chenyang Wu, Qian Chen, Akang Wang, Tian Ding, Ruoyu Sun, Wenguo Yang, and Qingjiang Shi. On representing convex quadratically constrained quadratic programs via graph neural networks, 2024.

Table 1: Taxonomy of closely related learning-to-optimize paradigms.

Paradigm	Representative coverage	Structure not covered
Sparse unrolling	LISTA, HyperLISTA, graph sparse-coding unrolling	Arbitrary matrix-defined LASSO graphs with size transfer
WL-based GNNs	LP, QP, QCQP, and SOCP expressivity	Algorithmic primal-dual trajectories and solver warm starts
AU-based L2O	PDHG-Net for LPs and PDQP-Net for QPs	Explicit SOC projection and cone-block message passing
Conic first-order solvers	PDHG-based matrix-free conic solvers	Learned graph/channel parameterization across instances
General constrained GNNs	Dual-ascent unrolled GNNs	SOCP-specific PDHG updates and cone geometry

- [20] Yiheng Xiao and Huikang Liu. On the convergence of doubly stochastic primal-dual hybrid gradient method. Manuscript, 2026.
- [21] Linxin Yang, Bingheng Li, Tian Ding, Jianghua Wu, Akang Wang, Yuyi Wang, Jiliang Tang, Ruoyu Sun, and Xiaodong Luo. An efficient unsupervised framework for convex quadratic programs via deep unrolling, 2024.